HOT WET EQUATORIAL CLIMATE

EQUATORIAL CLIMATE MAP → CONVECTION DAY CYCLE → TEMPERATURE-RANGE FIREWALL → RAINFALL IDENTITY → RAINFOREST VERTICAL PROFILE → NUTRIENT-CYCLE LOOP

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◈ SUBORDINATE EXTRA ★ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 EQUATORIAL CLIMATE MAP

EQUATORIAL CLIMATE MAP · PERSISTENT HIGH SOLAR RECEIPT · AMAZON + CONGO · MAJOR CONTINENTAL LOWLANDS · MALAYSIA + EAST INDIES · MARITIME EXPRESSION · ALTITUDE OR MONSOON MODIFICATION

EQUATOR → persistent high solar receipt

AMAZON + CONGO → major continental lowlands → MALAYSIA + EAST INDIES → maritime expression → HIGHLANDS / MARGINS → altitude or monsoon modification

SYSTEM / DEFINITION
- EQUATOR → persistent high solar receipt
- AMAZON + CONGO → major continental lowlands

PROCESS / MECHANISM
- MALAYSIA + EAST INDIES → maritime expression
- HIGHLANDS / MARGINS → altitude or monsoon modification

SPATIAL / INDIA PATTERN
- EQUATOR → persistent high solar receipt
- AMAZON + CONGO → major continental lowlands

HAZARD / LIMIT / RESPONSE
- MALAYSIA + EAST INDIES → maritime expression
- HIGHLANDS / MARGINS → altitude or monsoon modification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HIGHLANDS / MARGINS → altitude or monsoon modification.

01

STAGE 01 CONVECTION DAY CYCLE

CONVECTION DAY CYCLE · MORNING SUN · MOIST SURFACE HEATING · NOON ASCENT · EXPANDING AND COOLING AIR · CUMULONIMBUS THUNDERSTORM · RAIN-COOLED AIR; CYCLE CAN RECUR

MORNING SUN → moist surface heating → NOON ASCENT → expanding and cooling air → AFTERNOON → cumulonimbus thunderstorm → EVENING → rain-cooled air; cycle can recur

DEFINITION / COMPONENTS
- MORNING SUN → moist surface heating
- NOON ASCENT → expanding and cooling air

TRIGGER / MECHANISM
- AFTERNOON → cumulonimbus thunderstorm
- EVENING → rain-cooled air; cycle can recur

EFFECT / INTERVENTION
- MORNING SUN → moist surface heating
- NOON ASCENT → expanding and cooling air

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AFTERNOON → cumulonimbus thunderstorm; EVENING → rain-cooled air; cycle can recur.

02

STAGE 02 TEMPERATURE-RANGE FIREWALL

TEMPERATURE-RANGE FIREWALL · ANNUAL RANGE · USUALLY VERY SMALL · DIURNAL RANGE · LARGER THAN ANNUAL RANGE · WEAK SEASONALITY NEAR EQUATOR · MEANS DO NOT CAP DAILY EXTREMES

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
ANNUAL RANGE → usually very small	CAUSE → weak seasonality near Equator	ANNUAL RANGE → usually very small
DIURNAL RANGE → larger than annual range	CAUTION → means do not cap daily extremes	DIURNAL RANGE → larger than annual range

DEFINITION / COMPONENTS
- ANNUAL RANGE → usually very small
- DIURNAL RANGE → larger than annual range

TRIGGER / MECHANISM
- CAUSE → weak seasonality near Equator
- CAUTION → means do not cap daily extremes

EFFECT / INTERVENTION
- ANNUAL RANGE → usually very small
- DIURNAL RANGE → larger than annual range

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CAUTION → means do not cap daily extremes.

03

STAGE 03 RAINFALL IDENTITY

RAINFALL IDENTITY · RAIN ALL MONTHS · SUPPORTS AF READING · TRUE DRY SEASON · TEST AM OR AW INSTEAD · SOURCE NOTES OFTEN 150-250 CM · DISTRIBUTION OUTRANKS ONE ANNUAL NUMBER

RAIN ALL MONTHS → supports Af reading

TRUE DRY SEASON → test Am or Aw instead → TOTAL → source notes often 150-250 cm → RULE → distribution outranks one annual number

DEFINITION / COMPONENTS
- RAIN ALL MONTHS → supports Af reading
- TRUE DRY SEASON → test Am or Aw instead

TRIGGER / MECHANISM
- TOTAL → source notes often 150-250 cm
- RULE → distribution outranks one annual number

EFFECT / INTERVENTION
- RAIN ALL MONTHS → supports Af reading
- TRUE DRY SEASON → test Am or Aw instead

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRUE DRY SEASON → test Am or Aw instead.

04

STAGE 04 RAINFOREST VERTICAL PROFILE

RAINFOREST VERTICAL PROFILE · CROWNS ABOVE MAIN ROOF · DENSE LIGHT-CAPTURING LAYER · SHADE-TOLERANT GROWTH · DARK, HUMID, RAPID DECAY

EMERGENTS → crowns above main roof

CANOPY → dense light-capturing layer → UNDERSTOREY → shade-tolerant growth → FLOOR → dark, humid, rapid decay

SYSTEM / DEFINITION
- EMERGENTS → crowns above main roof
- CANOPY → dense light-capturing layer

PROCESS / MECHANISM
- UNDERSTOREY → shade-tolerant growth
- FLOOR → dark, humid, rapid decay

SPATIAL / INDIA PATTERN
- EMERGENTS → crowns above main roof
- CANOPY → dense light-capturing layer

HAZARD / LIMIT / RESPONSE
- UNDERSTOREY → shade-tolerant growth
- FLOOR → dark, humid, rapid decay

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UNDERSTOREY → shade-tolerant growth; FLOOR → dark, humid, rapid decay.

05

STAGE 05 NUTRIENT-CYCLE LOOP

NUTRIENT-CYCLE LOOP · RAPID WARM-WET DECOMPOSITION · ROOT MAT · RAPID NUTRIENT UPTAKE · PRINCIPAL NUTRIENT CAPITAL · HEAVY RAIN · LEACHING AFTER CLEARANCE

LITTER → rapid warm-wet decomposition → ROOT MAT → rapid nutrient uptake → BIOMASS → principal nutrient capital → HEAVY RAIN → leaching after clearance

DEFINITION / COMPONENTS
- LITTER → rapid warm-wet decomposition
- ROOT MAT → rapid nutrient uptake

TRIGGER / MECHANISM
- BIOMASS → principal nutrient capital
- HEAVY RAIN → leaching after clearance

EFFECT / INTERVENTION
- LITTER → rapid warm-wet decomposition
- ROOT MAT → rapid nutrient uptake

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BIOMASS → principal nutrient capital; HEAVY RAIN → leaching after clearance.

06

STAGE 06 FALLOW DECISION

FALLOW DECISION · SMALL CLEARING + LONG FALLOW · SHORTENED FALLOW · DECLINING BIOMASS · STEEP SLOPE + REPEATED FIRE · EROSION RISK · TENURE + LIVELIHOOD · MANAGEMENT OUTCOME

SMALL CLEARING + LONG FALLOW → recovery

SHORTENED FALLOW → declining biomass → STEEP SLOPE + REPEATED FIRE → erosion risk → TENURE + LIVELIHOOD → management outcome

DEFINITION / COMPONENTS
- SMALL CLEARING + LONG FALLOW → recovery
- SHORTENED FALLOW → declining biomass

TRIGGER / MECHANISM
- STEEP SLOPE + REPEATED FIRE → erosion risk
- TENURE + LIVELIHOOD → management outcome

EFFECT / INTERVENTION
- SMALL CLEARING + LONG FALLOW → recovery
- SHORTENED FALLOW → declining biomass

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STEEP SLOPE + REPEATED FIRE → erosion risk.

07

STAGE 07 HUMID-TROPICS ECONOMY

HUMID-TROPICS ECONOMY · CAPITAL + LABOUR + EXPORT CROP · MIXED FOREST · SELECTIVE LOGGING DIFFICULTY · ROAD ACCESS · EXTRACTION GEOGRAPHY · CLIMATE ENABLES; INSTITUTIONS DECIDE

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
PLANTATION → capital + labour + export crop	RIVER / ROAD ACCESS → extraction geography	PLANTATION → capital + labour + export crop
MIXED FOREST → selective logging difficulty	RULE → climate enables; institutions decide	MIXED FOREST → selective logging difficulty

DEFINITION / COMPONENTS
- PLANTATION → capital + labour + export crop
- MIXED FOREST → selective logging difficulty

TRIGGER / MECHANISM
- RIVER / ROAD ACCESS → extraction geography
- RULE → climate enables; institutions decide

EFFECT / INTERVENTION
- PLANTATION → capital + labour + export crop
- MIXED FOREST → selective logging difficulty

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PLANTATION → capital + labour + export crop.

08

STAGE 08 INDIA ANALOGUE MAP

INDIA ANALOGUE MAP · WESTERN GHATS · WINDWARD EVERGREEN BELT · WET EVERGREEN · EASTERN HIMALAYAN FOOTHILLS · INSULAR RAINFOREST

WESTERN GHATS → windward evergreen belt

NORTH-EAST → wet evergreen / semi-evergreen → EASTERN HIMALAYAN FOOTHILLS → transition → ANDAMAN-NICOBAR → insular rainforest

SYSTEM / DEFINITION
- WESTERN GHATS → windward evergreen belt
- NORTH-EAST → wet evergreen / semi-evergreen

PROCESS / MECHANISM
- EASTERN HIMALAYAN FOOTHILLS → transition
- ANDAMAN-NICOBAR → insular rainforest

SPATIAL / INDIA PATTERN
- WESTERN GHATS → windward evergreen belt
- NORTH-EAST → wet evergreen / semi-evergreen

HAZARD / LIMIT / RESPONSE
- EASTERN HIMALAYAN FOOTHILLS → transition
- ANDAMAN-NICOBAR → insular rainforest

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EASTERN HIMALAYAN FOOTHILLS → transition; ANDAMAN-NICOBAR → insular rainforest.

09

STAGE 09 OFFICIAL EVIDENCE FIREWALL

OFFICIAL EVIDENCE FIREWALL · FSI REPORT · DATED BIENNIAL ASSESSMENT · FOREST COVER · CANOPY-BASED MAPPED CLASS · FOREST TYPE · ECOLOGICAL CLASSIFICATION · LEGAL STATUS

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
FSI REPORT → dated biennial assessment	FOREST TYPE → ecological classification	FSI REPORT → dated biennial assessment
FOREST COVER → canopy-based mapped class	LEGAL STATUS → separate administrative category	FOREST COVER → canopy-based mapped class

DEFINITION / COMPONENTS
- FSI REPORT → dated biennial assessment
- FOREST COVER → canopy-based mapped class

TRIGGER / MECHANISM
- FOREST TYPE → ecological classification
- LEGAL STATUS → separate administrative category

EFFECT / INTERVENTION
- FSI REPORT → dated biennial assessment
- FOREST COVER → canopy-based mapped class

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FOREST TYPE → ecological classification; LEGAL STATUS → separate administrative category.

10

STAGE 10 VERIFIED PYQ PAIR

VERIFIED PYQ PAIR · RAINFOREST VEGETATION STRUCTURE · NUTRIENTS AND DECOMPOSITION · OFFICIAL ANSWER LETTERS WITHHELD · TEACH DISTINCTION; DO NOT INVENT KEY

CLASSIFICATION	DIAGNOSTIC BASIS	PROCESS LINK	EXAM TRAP
2021 → rainforest vegetation structure	LEDGER → official answer letters withheld	2021 → rainforest vegetation structure	LEDGER → official answer letters withheld
2023 → nutrients and decomposition	METHOD → teach distinction; do not invent key	2023 → nutrients and decomposition	METHOD → teach distinction; do not invent key

CLASSIFICATION
- 2021 → rainforest vegetation structure
- 2023 → nutrients and decomposition

DIAGNOSTIC BASIS
- LEDGER → official answer letters withheld
- METHOD → teach distinction; do not invent key

PROCESS LINK
- 2021 → rainforest vegetation structure
- 2023 → nutrients and decomposition

EXAM TRAP
- LEDGER → official answer letters withheld
- METHOD → teach distinction; do not invent key

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: METHOD → teach distinction; do not invent key.

11

STAGE 11 EQUATORIAL ANSWER SPINE

EQUATORIAL ANSWER SPINE · EQUATORIAL LOWLAND BELT · HEAT, CONVERGENCE, CONVECTION, RAIN · LAYERS, NUTRIENTS AND LAND USE · INDIA ANALOGUE AND SOURCE LIMITS

LOCATE → equatorial lowland belt → DERIVE → heat, convergence, convection, rain → LINK → layers, nutrients and land use → QUALIFY → India analogue and source limits

DEFINITION / COMPONENTS
- LOCATE → equatorial lowland belt
- DERIVE → heat, convergence, convection, rain

TRIGGER / MECHANISM
- LINK → layers, nutrients and land use
- QUALIFY → India analogue and source limits

EFFECT / INTERVENTION
- LOCATE → equatorial lowland belt
- DERIVE → heat, convergence, convection, rain

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LINK → layers, nutrients and land use; QUALIFY → India analogue and source limits.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · DO NOT GENERALISE · MOST INDIAN EVERGREEN BELTS ARE MONSOONAL, WHILE LIMITED AF · INDIA'S CLOSEST EQUIVALENT IS TROPICAL WET EVERGREEN FOREST IN · MAIN REGIONS · WINDWARD WESTERN GHATS · MALABAR , NORTH-EAST INDIA , ANDAMAN & NICOBAR ISLANDS · MALABAR VEGETATION RANGES FROM MOIST TROPICAL EVERGREEN TO MIXED

OPTIONAL PROCESS DEPTH
- Do not generalise: most Indian evergreen belts are monsoonal, while limited Af classification depends on the map/dataset.
- India's closest equivalent is tropical wet evergreen forest in very high-rainfall belts.
- Main regions: windward Western Ghats / Malabar , North-East India , Andaman & Nicobar Islands .
- Malabar vegetation ranges from moist tropical evergreen to mixed broad-leaved and monsoon deciduous types.

DATA / INSTITUTION LIMIT
- Andaman-Nicobar has tropical evergreen/semi-evergreen and littoral forests with high endemism.
- The archipelago combines forest and reef biodiversity with high seismicity and Barren Island volcanism.
- Western Ghats + North-East India are biodiversity-hotspot landscapes.
- Jhum is shifting cultivation; degradation rises when fallow cycles shorten.

CONTEMPORARY PARALLEL
- NOT: India has a true equatorial Af climate in the south → India lacks true Af; it has equatorial-type evergreen forests in wet monsoon belts.
- NOT: The leeward Deccan side of the Western Ghats has evergreen rainforest → Evergreen forest is on the windward western slope.
- the leeward side is drier.
- NOT: Andaman forests are ordinary dry deciduous forests → A&N contains tropical evergreen,

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels